REACTION WHEEL ACTUATOR DEVELOPMENT: JUMPING FROM HUNDREDS TO THOUSANDS

A. Wade[(1)], R. Jackson[(1)], J. Keeble[(1)], J. Orr[(1)]

[(1)] Rocket Lab, 25 Levene Place, Mount Wellington, Auckland, New Zealand, Email:a.wade@rocketlab.co.nz

ABSTRACT

Sinclair Interplanetary by Rocket Lab has delivered more than 360 reaction wheel actuators to orbit. The available products range from a minuscule 0.003Nms wheel for Cubesats to a 0.1Nm 1Nms reaction wheel suitable for small spacecraft. This paper covers the development of a new 0.2Nm 12Nms reaction wheel by Rocket Lab ideal for use in telecommunication satellites and the production line required to produce 1,000's of units per year.

The 12Nms reaction wheel actuator product has been developed from a Sinclair Interplanetary heritage design and incorporates design for manufacturing features to make the reaction wheel easier to assemble. A succession of development models has been tested in representative environments including vacuum and vibration.

Rocket Lab have also tackled the challenge of designing a production line capable of producing 1,000's of 12Nms reaction wheels a year. The reaction wheel has been designed with injection moulded parts, limited adhesives, and no harnessing to reduce the assembly time. Multiple automated stations are used to burn in the electronics, assemble the rotors, wind the stators, and test the final assembled product.

The result of this development is a robust RW5.12 reaction wheel actuator developed for use in low earth orbit environments, and a RW4.12 reaction wheel actuator that has been developed for use in higher radiation environments. The electronics architecture is different for each variant, but the mechanical parts can be produced on the same production line. The first flight units of the RW5.12 will be delivered in 2023.

1 INTRODUCTION

Rocket Lab has been providing frequent and reliable access to space since 2017 with the Electron launch vehicle. Rocket Lab also manufactures satellites and spacecraft components that give customers the freedom to design missions their own way, spend less, and reach orbit faster. Rocket Lab supports the space ecosystem as a merchant supplier offering a broad set of space proven component products, having flown on over 1,700 missions to date.

Spacecraft components, many based on acquisitions by Rocket Lab, are offered at a manufacturing scale capable of suppling constellation-class projects by fulfilling large volume deliveries. This capability has already been demonstrated on multiple programs to date. Spacecraft component offerings include high performance star trackers and reaction wheel actuators from Sinclair Interplanetary; the MAX flight and ground software suite, high performance GNC software, and constellation autonomy/management software from Advanced Solutions Inc.; launch separation systems from Planetary Systems Corp.; solar cells, solar panels, and solar arrays from SolAero Technologies Corp., and deep-space capable radios licensed from the Johns Hopkins University Applied Physics Laboratory.

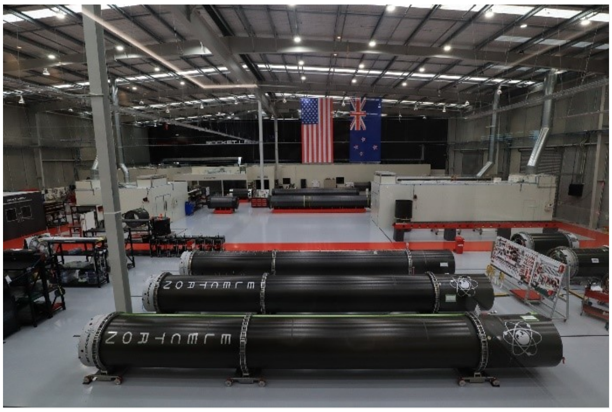

Figure 1 Rocket Lab production facility in New Zealand

2 OVERVIEW

The 12Nms product range currently has two variants: RW5.12 and RW4.12. The RW5.12 variant is designed to operate on a 30-50V bus using CAN communications. The components used in the design allow for operation in a low earth orbit radiation environment with a total ionising dose of 12kRads.

The RW4.12 variant is designed to operate on a 24-34V bus using RS485 communications. The electronics are designed to be more tolerant of higher orbits with >60krad total ionising dose possible.

Both variants use a common mechanical design shown in Figure 2 with hybrid ceramic bearings, space qualified grease lubrication and a fully demisable rotor. The reaction wheels can be operated in momentum or torque control mode and can provide several health monitoring telemetry points.

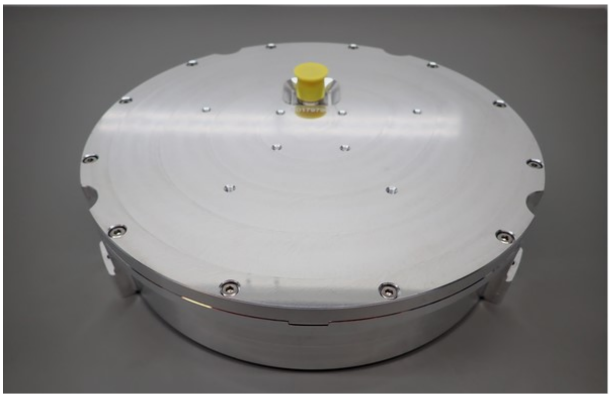

Figure 2 12Nms Reaction Wheel Actuator

Figure 3 contains the torque-momentum performance measured during a full motor current test at the temperature extremes for different operating voltages after environmental qualification.

3 CRITICAL DESIGN

The design for the 12Nms product range is built on Sinclair Interplanetary heritage which has been enhanced by incorporating techniques for mass production. Figure 4 shows an exploded view of the RW5.12 which highlights the main subassemblies.

Where possible high volume manufacturing processes have been used. Variable processes such as the use of adhesives, thread locking and harnessing have been minimised or eliminated. Self-locating features are used to avoid awkward assembly processes that are dependent on high levels of experience and skill.

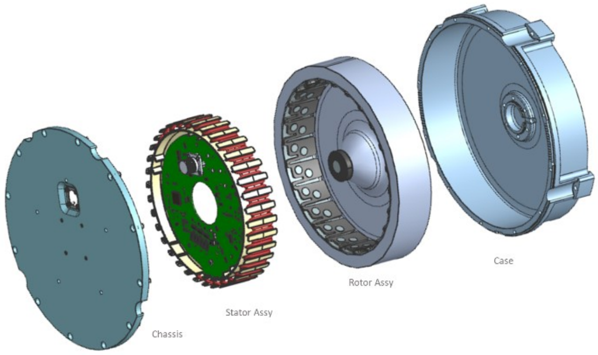

Figure 4 Exploded view of RW5.12 assembly

3.1 Rotor Assembly

The rotor is primarily made of aluminium to ensure that the rotor is fully demisable. The challenge presented by this approach is the volume of aluminium required to achieve the correct momentum. Power constraints prevent the reaction wheel from rotating any faster to compensate for the lower inertia of an aluminium rotor.

A pancake style motor has been used so that the active components of the motor can usefully contribute to the inertia of the rotor but an increase in diameter is unavoidable to ensure the reaction wheel is fully demisable.

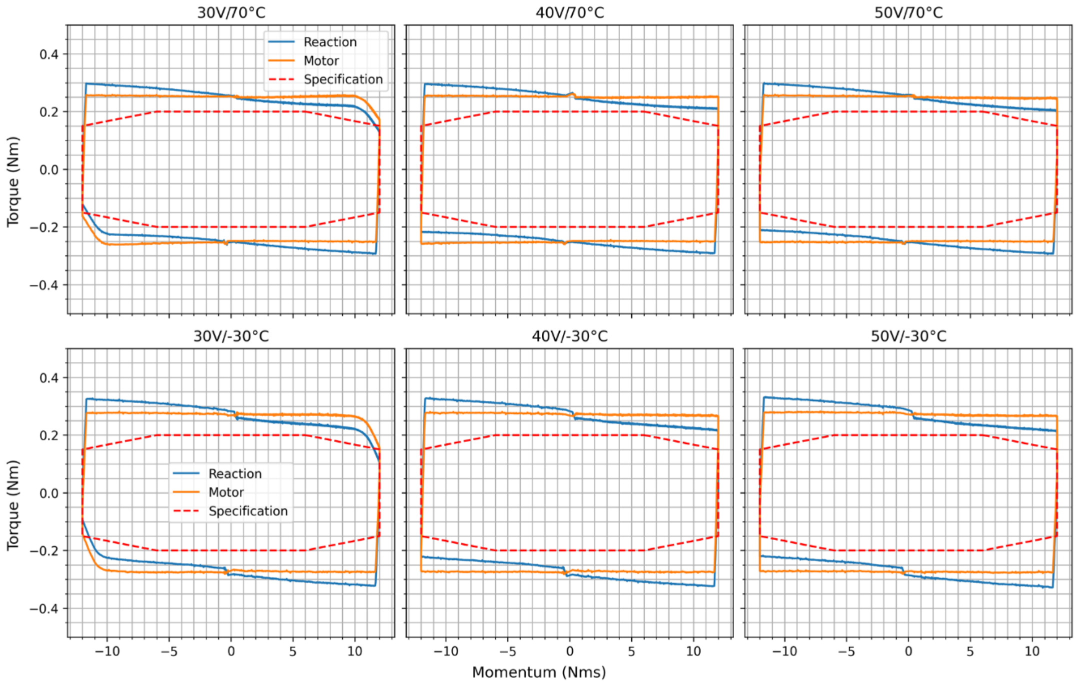

Figure 3 RW5.12 Reaction Torque Performance

3.2 Stator Assembly

Internal harnessing has been eliminated by arranging the internal structure of the reaction wheel such that the connectors directly interface with the electronics. The stator frame includes features for interfacing with the structure, electronics, coil assembly, wire separation and hall sensor alignment.

The stator frame is manufactured using injection moulding which enables this complex part to be produced at a lower cost. The final parts have been tested to ensure outgassing limits would not be exceeded, and a thermal gasket is used to transfer heat from the motor coils to the chassis.

3.3 Housing

The design of the housing is critical as it controls the alignment of the bearings. Fasteners have been minimised and adhesives have been replaced with locking threaded inserts to reduce assembly time and eliminate process inconsistency. Tight control of locating features allows for the rotor to be balanced separately before integration but allow the reaction wheel to meet unbalance requirements post vibration.

4 DEVELOPMENT TESTING

Rocket Lab can bring products to market quickly because of a strong focus on early testing. The prototype mechanical design for the RW5.12 has been through many iterations and was thoroughly tested in vibration and thermal vacuum prior to the formal qualification. This allows for all major design risks to be retired through testing during the critical design phase rather than relying on results from models using large design margins.

4.1 Motor Design Iteration

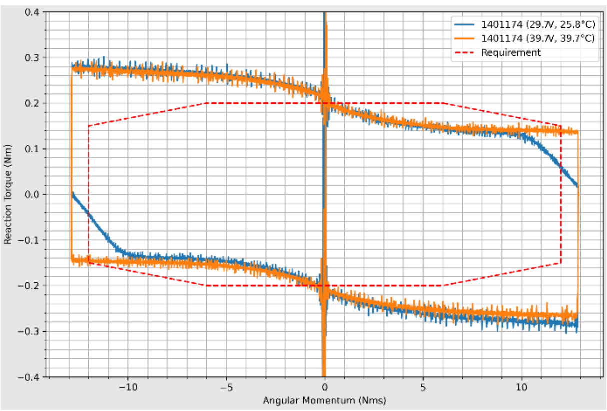

Figure 5 Reaction torque measurement for RW5.12 prototype during TVAC campaign

Figure 5 shows the result from the first round of development testing in thermal vacuum which showed higher viscous drag in the bearings than anticipated. A new prototype with a different motor winding was created. The new motor design increased available torque at the cost of speed, so the rotor inertia was increased to compensate.

4.2 Housing Design Iteration

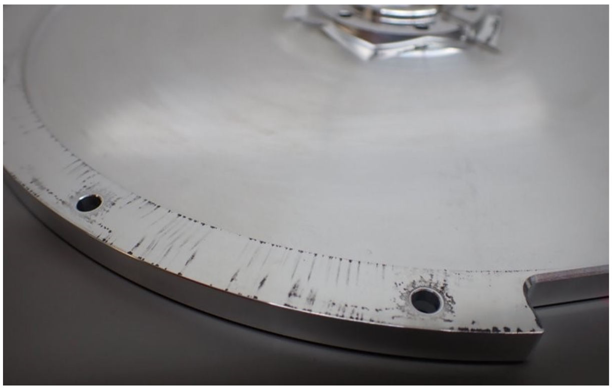

Figure 6 Excessive fretting damage seen during vibration testing of prototype design

Development testing included multiple vibration campaigns which included quasi-static and random vibration. Shock was not included at this stage due to test equipment limitations. Special units which allowed the measurement of the rotor response were used to evaluate the load that bearings experienced during vibration.

Figure 6 shows that excessive fretting was seen at some interfaces after a prototype unit was subjected to vibration. The mechanical enclosure was changed to reduce fretting by shifting the housing interface to a stiffer part of the housing which eliminated the fretting in following vibration testing.

4.3 Bearing Performance

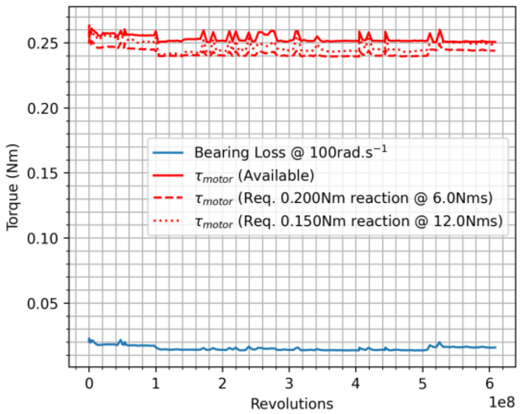

Figure 7 Life derisking of final prototype design in-air

The final prototype design was subjected to the appropriate quasi-static and random vibration levels before another thermal vacuum campaign. The prototype is currently undergoing a life derisking activity. This involves operating the reaction wheel at high temperature whilst rotating at high speed to derisk any bearing damage/fatigue issues after vibration.

Figure 7 shows the unit has successfully completed >600 million revolutions in June 2023 with bearing inspection planned once the unit reaches 1 billion revolutions. This concluded the development phase of the project and allowed for the completion of the critical design review with the lead customer.

5 QUALIFICATION

The RW5.12 successfully completed qualification in June 2023 with a follow up qualification programme planned for the RW4.12 by the end of 2023. This section provides an overview of the key sections of the qualification campaign.

5.1 Performance Verification

The first stage of the qualification was to verify that the reaction wheel had been built to specification and passed all the acceptance testing that would normally be performed prior to delivery. This testing involves burn-in whilst thermally cycling the electronics over a two week period with dynamic testing of the reaction wheel as the final step to proving the reaction wheel meets the design specification.

5.2 Static and Dynamic Unbalance Measurement

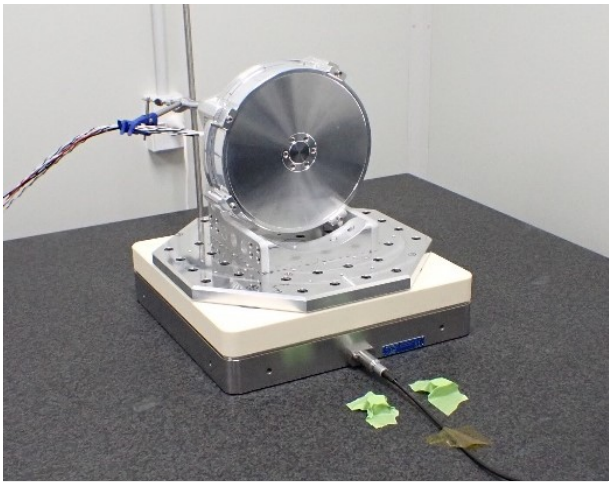

Figure 8 Qualification unit during unbalance measurements

Figure 8 shows that before environmental testing began the reaction wheel was mounted to Rocket Lab's microvibration dynamometer to measure the as-assembled static and dynamic unbalance. The exported forces were measured with the reaction wheel fitted directly to the table, and unbalance measurements are made with a special bracket to place the centre of mass of the rotor along the z-axis of the dynamometer. The measurement is taken before and after environmental testing to ensure the reaction wheel remains in specification.

5.3 Vibration

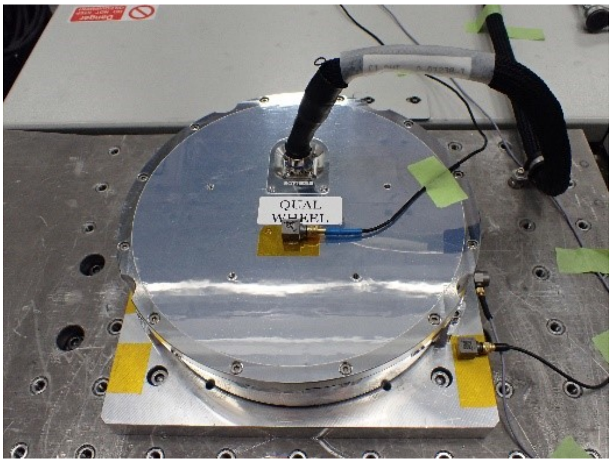

Figure 9 Qualification unit during vibration testing

Figure 9 shows the qualification unit during vibration testing simulating launch using in-house facilities at Rocket Lab. Each axis was subjected to a quasi-static sine vibration of 9g and random vibration to 14.8grms in each axis. The RW4.12 will follow the same methodology and will be qualified to 20g sine, and 20grms random vibration which has been proven to be survivable during development testing.

5.4 Shock

The qualification unit was shocked at external facilities in New Zealand to 1414g above 1000Hz. The RW4.12 will follow the same methodology and will be qualified to 800g above 1000Hz. Shock testing was the last qualification activity which had not been previously de-risked as part of the development activities.

5.5 Thermal Vacuum

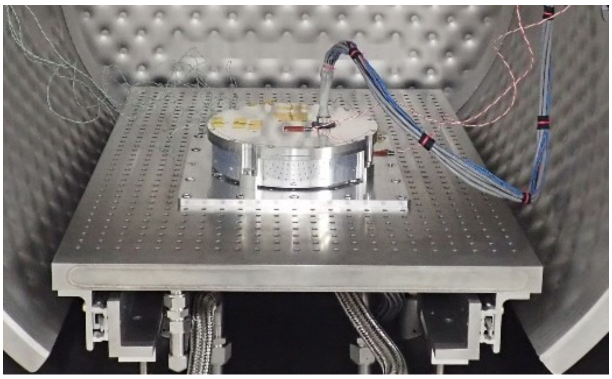

Figure 10 Qualification unit in TVAC

Figure 10 shows the qualification unit was thermally cycled in a vacuum chamber. The testing included one survival cycle to the non-operational qualification temperatures followed by 28 thermal cycles.

Performance at temperature extremes was characterised on the first and final thermal cycle. The remainder of the testing involved exercising the reaction wheel by following a low duty sinusoidal momentum profile to simulate typical variation seen during orbital operations.

5.6 Post-Environmental Testing

The performance verification and unbalance measurements were repeated at the conclusion of the environmental testing for comparison.

5.7 Disassembly and Inspection

Figure 11 shows the final stage of the qualification which involved dismantling the reaction wheel for inspection to determine if any foreign object debris had been generated, that the electronics were in good condition and that the floating bearing interface was still able to slide.

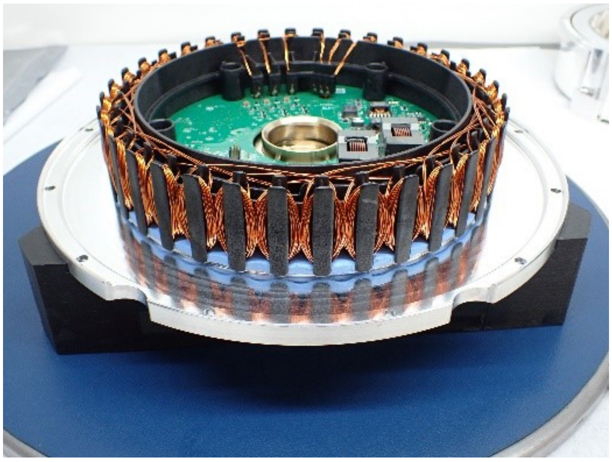

Figure 11 Qualification unit being inspected post environmental testing

5.8 Results

The results presented in Table 1 and Figure 3 are from the final thermal cycle during thermal vacuum testing which show the qualification unit was still performing correctly after environmental testing. The reaction wheel unbalance was still well within specification post-vibration which was a critical result for the development.

No foreign debris or issues were found during disassembly and inspection. The RW5.12 product has passed qualification to the lead customer requirements which gives high confidence that the RW4.12 will pass qualification in 2023 as only the electronics are different.

Table 1 Performance measurements of RW5.12 post environmental qualification

Characteristic	Unit	Value
Reaction Torque	Nm	@6Nms: 0.2 @12Nms: 0.15
Momentum	Nms	±12
Temperature (Operational)	°C	-20 to 60
Temperature (Non-Operational)	°C	-40 to 70
Power Consumption (0.2Nm, max)	W	@70°C: 160 @-30°C: 160
Power Regeneration (-0.2Nm, max)	W	@70°C: 70 @-30°C: 70
Power Consumption (0Nm, 12Nms)	W	@70°C: 30 @-30°C: 36
Static Unbalance	g.mm	<26
Dynamic Unbalance	$g.mm^2$	<800
Mass	kg	<4.7
Envelope	mm	Ø264 x 63

6 PRODUCTION

Manufacturing the 12Nms reaction wheels required a new production line which was developed in tandem with the product design. The production line has been sized to produce 2000 wheels per annum, or 40 wheels per week on a single-piece flow basis. While these volumes are high-volume in the space sector, this is considered medium volume in most manufacturing industries.

The production line design lends itself to a combination of semi-automated and manual tasks to achieve these volumes, depending on the skill/complexity of the task, the quality level required, the risk exposure created by a failed task, the time available, and the cost to complete the task.

Critical requirements of the product design have been determined by both design engineers and manufacturing engineers, and the checking of these key requirements have been automated where possible to ensure acceptable product progresses through each stage of the assembly.

Additional to the physical aspects of the new production line, associated production *systems* have also been designed alongside the line. These systems have focused on streamlining material supply to the line, fast identification and decision making related to non-conformances, and fast analysis of test results. The automated production stations have been set up to automatically pass test data to the production database for analysis and long-term storage.

Qualification of the production line has been undertaken in parallel with the product qualification previously described, and in accordance with AS9100 requirements. This ensures that the line can consistently produce

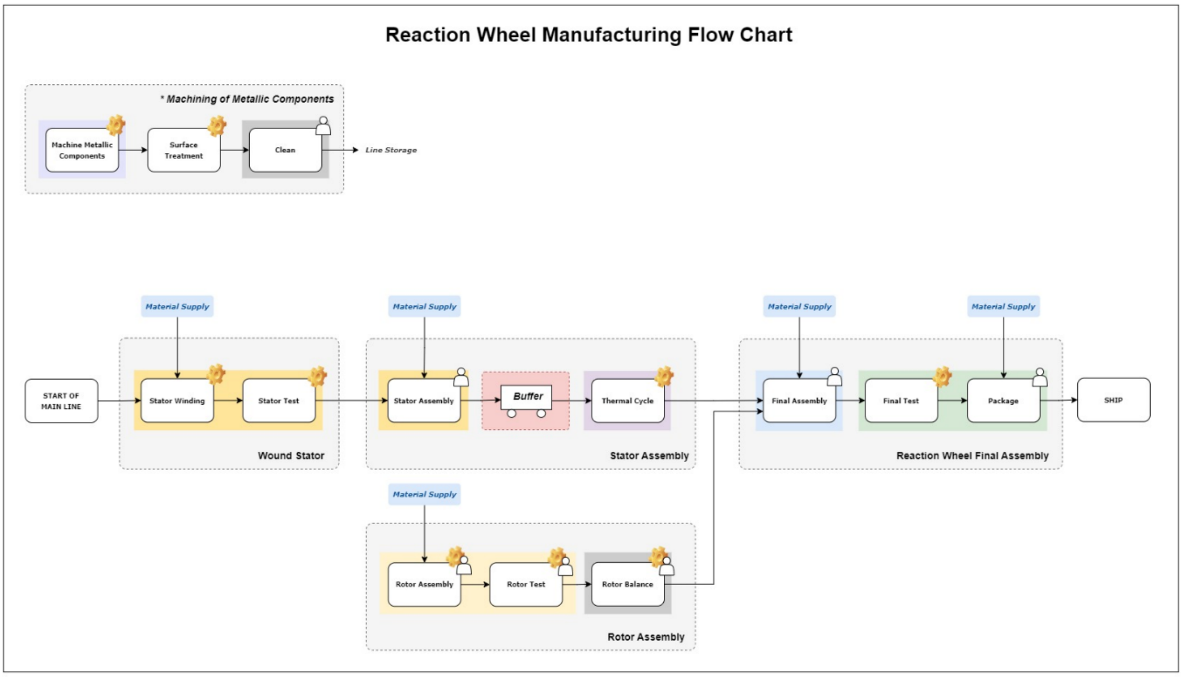

Figure 13 Reaction Wheel Production Line

product which meets specification. Figure 13 gives an overview of the manufacturing flow for the 12Nms reaction wheel.

6.1 Stator Assembly and Testing

Figure 12 Stator assembly during burn-in testing

The stator assembly includes the motor windings and drive electronics of the reaction wheel. The motor windings are completed with an automated machine before being tested and integrated with the electronics. Figure 12 shows stator assemblies being tested in a burn-in test which involves 14 thermal cycles and an elevated temperature dwell for a total of 200 hours. Functional testing is completed periodically during burn-in to ensure the electronics and motor windings are in good health.

6.2 Rotor Assembly and Balancing

Figure 14 shows the rotor assembly which includes the bearings and the passive motor components of the reaction wheel. Magnets are installed and tested using an automated station before the rotor is then balanced.

Figure 14 Rotor being assembled

Rotor balancing is one of the key challenges of the production line as it is critical for customer operations and to improve bearing life by reducing the loads acting on the bearing. While several balancing methods were considered, the final solution involves material removal from the rotor body.

This process was initially outsourced but moved in house to remove the associated supply chain risk. An automated piece of equipment has been developed which undertakes a measurement of unbalance, calculates the required material removal at discrete intervals, and undertakes this removal. The process can be iterated as required to achieve the required unbalance specification.

6.3 Final Assembly and Testing

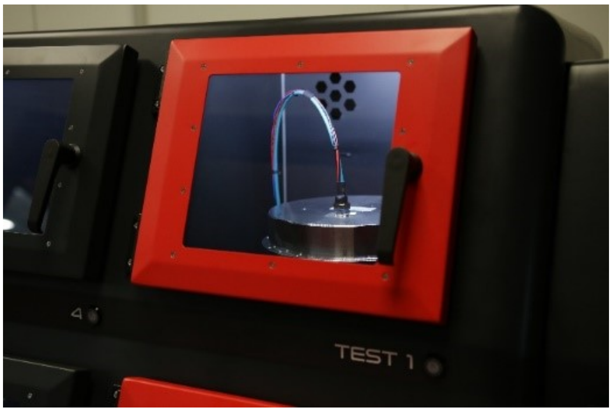

Figure 15 RW5.12 in final test station

The final step of the assembly process is to bring together the chassis, stator assembly and rotor assembly. The reaction wheel is operated at different speeds to distribute lubricant in the bearing before being moved to a final test station shown in Figure 15. The reaction wheel is automatically tested to check the motor parameters, bearing performance and motor torque to ensure the unit meets the specification before shipping to a customer.

7 QUALITY

Customers receiving multiples of a product inherently expect conformance to a specification and to present similar characteristics unit to unit. This becomes intrinsically more complicated when manufacturing en masse. Unit variation is undesired, and any rework to return a product to specification is expensive and disruptive to the production line.

The traditional approach to quality in the space industry is to verify the quality of a mechanism through extensive testing of every unit. Performing vibration and thermal vacuum testing on 1000's of units a year is prohibitively expensive and time consuming.

Another major expense is product assurance through inspection, where no value is directly added to the product, rather just a confirmation that the inputs to the process, and the output of the process were correct. If the manufacturing process can be validated through empirical evidence, with a statistically significant number of data points, the need for inspection and testing can be reduced.

An example of this approach can be seen with the metallic parts which are machined in-house on dedicated machines. Machined component production trials included 100% inspection which was completed to start collecting statistical evidence around process control and capability. This would allow for the collection of evidence enabling a move towards a sampling plan when in production. This can be done when the process is stable and showing minimal variability, typically assessed using the process capability metric, CpK.

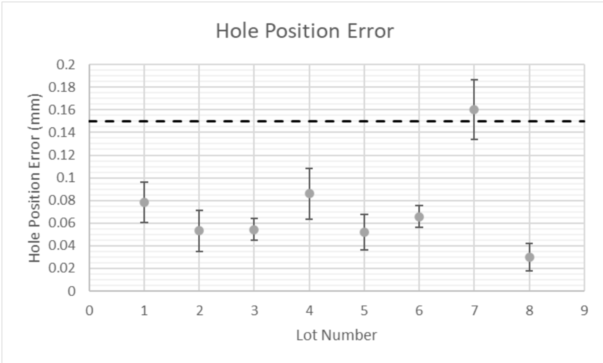

Figure 16 Hole position error

Figure 16 shows the result of measurements of the satellite mounting hole locations against production lot (sample size: 101 points). Lot 1 through 6 were completed during development, where it was thought the process was reasonably stable and capable, with a CpK of 1.04 so production commenced. The first production run (Lot 7) showed a shift in hole positional error due to machine setup inconsistencies. Verification did not exist within the manufacturing process until parts were fed down the production line for CMM inspection.

When dealing with medium-to-high volume manufacture, this could result in dozens of non-conforming parts being produced prior to the issue being found, resulting in large amounts of scrap and ultimately a line stoppage due to part shortages.

Mitigating this risk involved adding an additional verification during machine setup which provided consistent feedback to the technician that the correct offsets have been entered. Lot 8 shows that the average hole positional error reduced back down to a level consistent with those previously seen during development. Ongoing statistical process control methods have been employed to detect shifts or trends in the process.

8 CONCLUSION

Rocket Lab has developed and successfully qualified a 0.2Nm 12Nms reaction wheel actuator to add to its product range whilst designing a production line capable of producing 1000's of units per year to meet the demands of the current market.

A more radiation tolerant design for higher orbits is currently in development based on the RW5.12 qualified in this activity which is due to complete a qualification campaign towards the end of 2023. Table 2 contains the final specification for both products.

Table 2 RW4.12/RW5.12 Specification

Characteristic	Unit	RW5.12	RW4.12
Reaction Torque	Nm	@6Nms: 0.2 @12Nms: 0.15	
Momentum	Nms	±12	
Control Mode		Momentum or torque	
Supply Voltage	V	30-50	24-34
Power (0.2Nm, max)	W	<160	TBC
Power (0Nm, 12Nms)	W	<36	TBC
Thermal	°C	Survival: -40 to 70 Operational: -20 to 60	
Radiation	krad	>12	>60
Vibration	g_{rms}	In-plane: 14.1 Normal: 20	
Static Unbalance	g.mm	<26	
Dynamic Unbalance	$g.mm^2$	<800	
Communication	-	CAN-FD	RS485

8.1 Lessons Learned

Reaction wheels are difficult to develop as although they are conceptually simple, practically they are very difficult to get right. The additional challenge of designing a reaction wheel which can be produced easily in quantities of 1000's makes a difficult product seem impossible.

The main lesson learned during this project is that traditional mechanisms engineers in the space industry can struggle to transition to a mass manufacture mindset. The right skills from terrestrial industries are required to challenge design techniques, material selection, quality standards, and to design the infrastructure needed to manufacture at scale. The same transition occurred when the space industry moved from bespoke electronic components to using commercial-off-the-shelf electronics and qualifying them for use in space reducing the cost of access to space.

8.2 Future Work

Qualification has been completed which proves the reaction wheel will survive the launch environment and operating in space. However, the last remaining activity on the technical side is to perform a life de-risking activity.

Proving reaction wheel life is a tricky thing to get right. Current programmes tend to involve acceleration so that a certain revolution target can be completed in a representative way at increased speed by changing the viscosity of the lubricant to match nominal operating tribological conditions. However, these programmes cannot usefully accelerate the degradation, evaporation, or migration of the lubricant. There are also large uncertainties in the models used to determine what the nominal tribological conditions are [1].

The 12Nms product uses grease lubricated hybrid ceramic bearings. Adhesive wear is less of a concern in hybrid ceramic bearings and grease lubricated bearings demonstrate low speed elastohydrodynamic behaviour due to a thickener dominated lubricant in the contact [2]. This ensures bearing surfaces are separated even at low speed in grease lubricated bearings.

The 12Nms life derisking activity will measure the film thickness in the bearing using a specially instrumented reaction wheel which will allow Rocket Lab to identify the transition from the thickener dominated grease regime to the more desirable base oil dominated elastohydrodynamic regime. This will be performed across the operating temperature range so that customers can be provided with clear instructions on favourable operation speeds.

The information will be used to accelerate a derisking test where the reaction wheel is operated in the thickener dominated regime to prove the reaction wheel can operate in this life limiting operational area for the duration of the mission.

9 REFERENCES

1. Lewis, S, Buttery, M, Poyntz-Wright, O, Kent, A, Vortsellas, A (2018). "Accelerated Testing of Tribological Components - Uncertainties and Solutions", Proc. 44th Aerospace Mechanisms Symposium, NASA Glenn Research Center

2. Kanazawa, Y, Sayles, R.S, Kadiric, A (2017) "Film formation and friction in grease lubricated rolling-sliding non-conformal contacts", Tribology International, Volume 109, Pages 505-518, https://doi.org/10.1016/j.triboint.2017.01.026